

LAKE GRACE, Australia

WMO: 956370

Lat: 33.1006S

Lon: 118.4647E

Elev: 287

StdP: 97.93

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.0	3.2	-5.3	2.5	26.4	-3.2	3.0	23.6	10.7	15.0	9.5	15.1	0.8	0	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	37.4	18.4	35.1	18.3	33.1	17.8	26.0	26.7	21.7	27.1	19.7	27.4	5.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.8	26.4	20.5	15.7	22.4	17.7	13.2	21.4	82.1	26.6	64.7	27.4	57.6	27.6	36.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.1	8.7	7.8	-0.2	41.6	1.0	1.5	-1.0	42.7	-1.5	43.5	-2.1	44.3	-2.8	45.4
			-0.5	29.5	0.9	5.8	-1.2	33.7	-1.7	37.1	-2.2	40.3	-2.8	44.5

DB
WB

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	22.8	22.8	21.0	17.8	14.1	11.6	10.4	11.1	12.6	16.0	19.2	21.1
	DBStd	5.43	3.42	3.32	3.53	3.24	2.62	2.17	2.06	2.28	2.90	3.35	3.69	3.62
	HDD10.0	52	0	0	0	0	1	8	21	14	7	1	0	0
	HDD18.3	1183	3	3	13	47	134	203	246	226	174	89	33	12
	CDD10.0	2485	397	358	341	236	128	55	33	47	84	186	277	343
	CDD18.3	574	142	128	96	32	3	0	0	0	2	17	59	97
	CDH23.3	6853	1686	1402	979	302	26	1	0	3	46	295	818	1296
	CDH26.7	3237	886	697	424	86	3	0	0	9	104	368	660	
Wind	WSAvg	4.0	4.9	4.7	4.2	3.5	3.2	3.5	3.4	3.5	3.8	4.3	4.7	4.8
Precipitation	PrecAvg	345	19	19	22	23	43	49	45	41	31	20	22	13
	PrecMax	545	168	82	152	106	151	111	106	102	64	76	130	68
	PrecMin	196	0	0	0	0	3	8	13	5	5	2	0	0
	PrecStd	78	37	20	32	22	31	23	20	20	14	15	24	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	39.0	37.0	32.7	26.0	21.5	19.4	22.7	27.8	32.7	36.5	39.8
		MCWB	19.4	19.6	19.8	17.1	13.8	13.0	12.5	13.3	14.1	15.5	17.9	17.7
	2%	DB	37.1	35.9	34.1	29.0	23.5	19.4	17.6	19.9	24.1	29.8	33.7	36.3
		MCWB	18.3	18.7	19.2	17.0	13.8	12.2	11.9	12.5	13.3	16.1	16.7	17.6
	5%	DB	34.4	33.6	31.3	26.7	21.5	17.8	16.2	18.0	21.4	26.6	31.1	33.2
		MCWB	18.4	19.0	17.9	16.3	13.3	11.8	11.7	12.2	13.2	15.1	16.8	17.0
	10%	DB	31.7	31.0	28.6	24.5	19.7	16.3	15.0	16.4	19.2	23.9	28.1	30.3
		MCWB	17.9	18.1	17.5	15.6	13.1	11.6	11.1	11.6	12.4	14.0	15.7	16.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.0	28.9	29.5	26.0	18.8	16.8	15.9	17.0	19.4	24.2	29.4	29.1
		MCDB	29.6	29.5	29.9	26.4	20.8	17.0	16.6	17.5	19.8	24.6	30.2	30.1
	2%	WB	22.3	22.5	21.6	18.9	16.3	14.7	13.7	14.5	15.5	17.3	20.0	20.0
		MCDB	29.2	27.9	24.3	24.3	19.8	16.7	15.5	17.3	20.9	24.6	22.5	29.2
	5%	WB	20.2	20.3	19.4	17.5	15.1	13.5	12.6	13.3	14.1	15.7	17.5	18.4
		MCDB	29.0	28.5	26.3	23.5	18.8	15.9	14.7	16.5	19.6	24.0	26.7	28.8
	10%	WB	18.8	19.1	18.3	16.5	14.0	12.5	11.6	12.2	13.0	14.6	16.3	17.2
		MCDB	28.9	27.9	25.5	22.1	17.8	15.0	14.2	15.3	17.9	22.3	25.7	27.6
Mean Daily Temperature Range	5% DB	MDBR	15.1	14.2	12.8	11.5	10.7	9.7	9.4	10.4	12.2	13.5	14.5	15.1
		MCDBR	19.5	18.5	16.5	14.6	13.5	11.7	10.9	13.4	16.2	18.2	19.3	20.3
		MCWBR	5.9	5.6	6.1	6.0	6.3	6.5	6.6	7.5	7.4	7.3	6.9	6.0
	5% WB	MCDBR	15.9	15.3	13.1	11.9	9.7	9.1	9.1	10.3	13.3	15.2	15.9	16.9
		MCWBR	8.2	7.8	7.4	6.8	6.0	6.8	7.1	7.2	7.9	7.4	8.5	7.4
Clear-Sky Solar Irradiance	taub	0.336	0.330	0.316	0.305	0.291	0.285	0.286	0.290	0.310	0.318	0.322	0.325	
	taud	2.504	2.538	2.562	2.588	2.601	2.598	2.589	2.576	2.493	2.494	2.504	2.512	
	Ebn at Noon	1002	988	967	925	889	868	885	928	957	988	1009	1016	
	Edh at Noon	114	106	97	84	73	69	73	83	102	110	113	114	
All-Sky Solar Radiation	RadAvg	7.97	7.09	5.79	4.25	3.17	2.64	2.80	3.61	4.87	6.31	7.41	8.16	
	RadStd	0.47	0.37	0.31	0.27	0.21	0.12	0.15	0.15	0.21	0.36	0.44	0.44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	-2.27	-4.02	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page